

TELCO CUSTOMER CHURN

From Raw Data to Retention Strategy

A full analytical pipeline — from SQL exploration to an executive Power BI dashboard — built to answer one question: *where is this telecom losing customers, and what should it do next?*



CHURN SIGNAL — EARLY TENURE IS THE CRITICAL RETENTION WINDOW

STABILIZING AFTER INTERVENTION

7,043

CUSTOMERS ANALYZED

32

BUSINESS QUERIES

4

DASHBOARD PAGES

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Customer churn is not just a customer problem — it's a **revenue problem**.

01

Who is churning?

02

Why are they churning?

03

What should the business do about it?

Customer



Churn



Revenue Loss



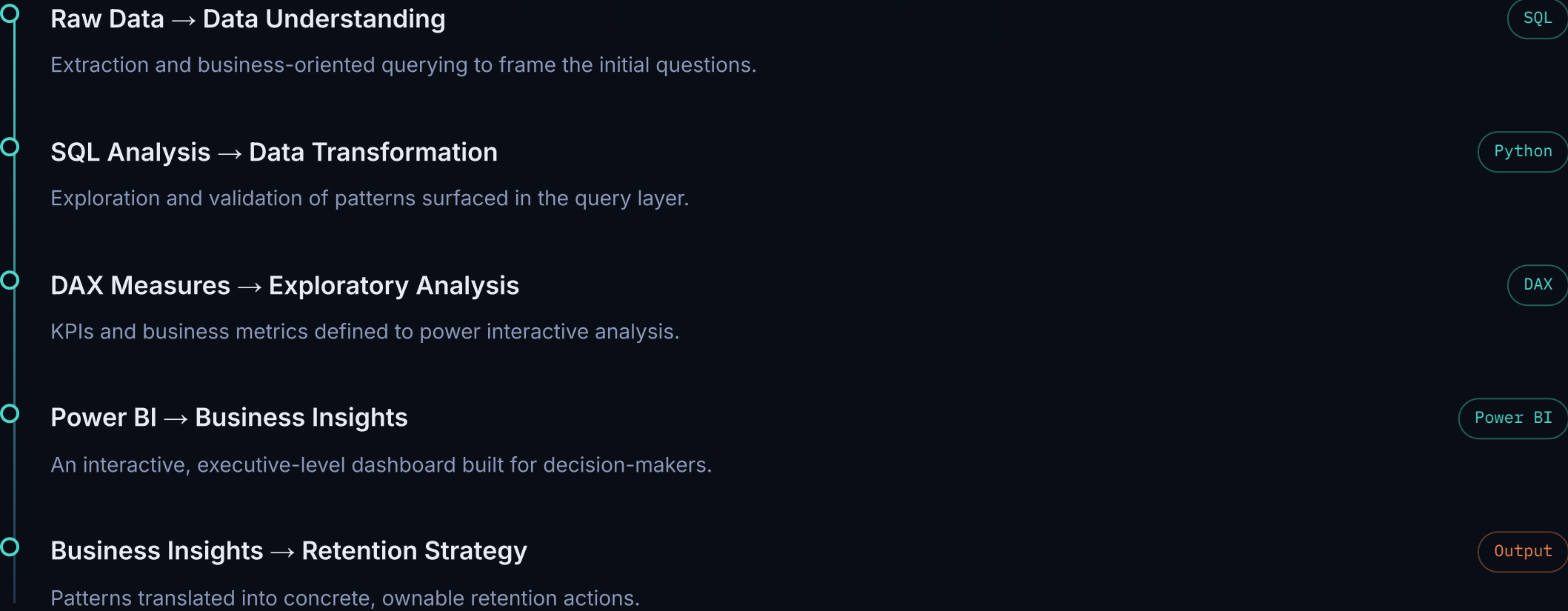
Retention Strategy

ANALYTICAL OBJECTIVE

Identify the customer segments, behavioral patterns, and service characteristics associated with churn — then translate the findings into actionable retention strategies.

A structured pipeline, not a single dashboard.

Every stage below fed the next — from raw records to a business recommendation the retention team could act on.



The dashboard is the output. This is the thinking behind it.

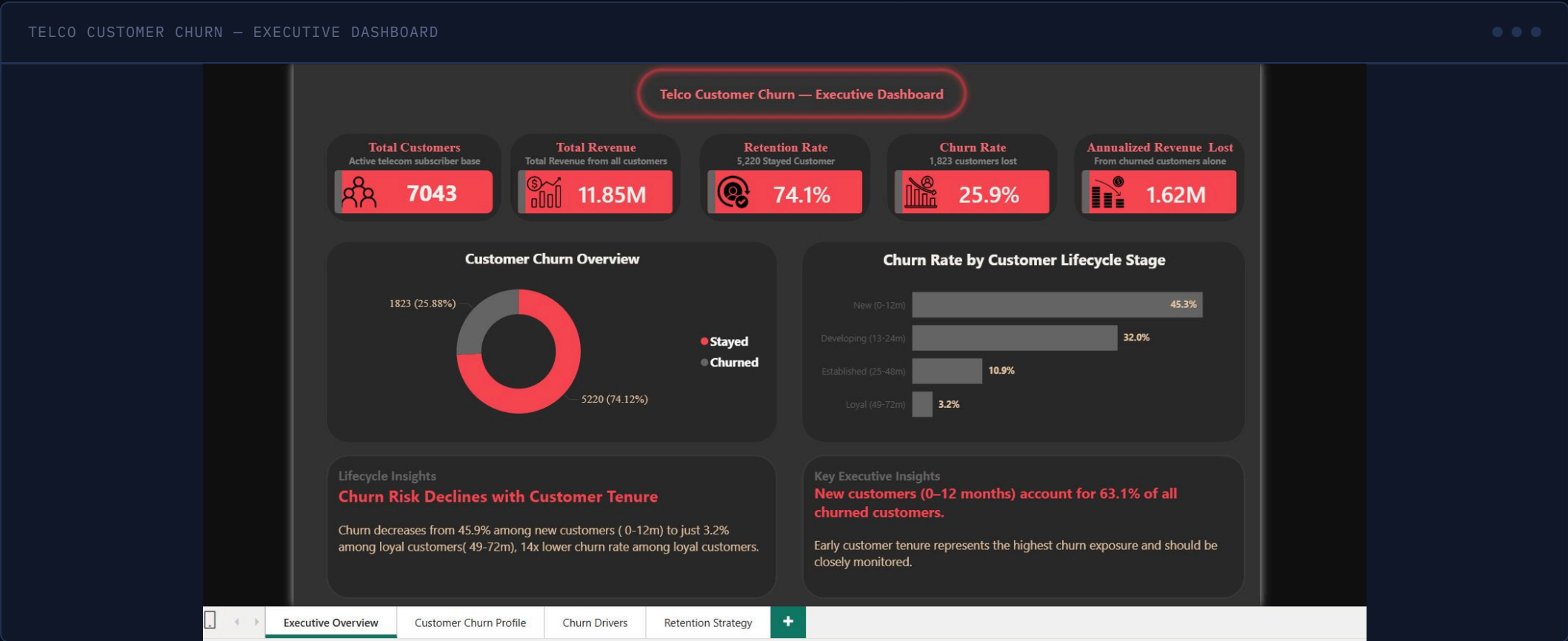
	PHASE 1 — BUSINESS CONTEXT
	The company provides internet service and phone service.
	Recently, customer churn rate increased. Management is worried because acquiring new customers is expensive, losing loyal customers reduces revenue, and churn affects long-term business growth.
	So we need to understand why customers leave.
	What's customer churn? A customer stops using the company's services — at a telecom, that means cancelling the subscription and stopping payment.

Phase 2 — Data Understanding
Mapped every table to a set of business questions before opening Power BI: customer table, service table, payment table, and demographics.
Questions I framed
Which service loses more customers — phone or internet? Does tenure affect the likelihood of staying? Which contract type retains customers best? Which family status churns more?
Payment & revenue
What's the average monthly charge? Which payment method is hardest for retention? What are the top churn reasons to act on?

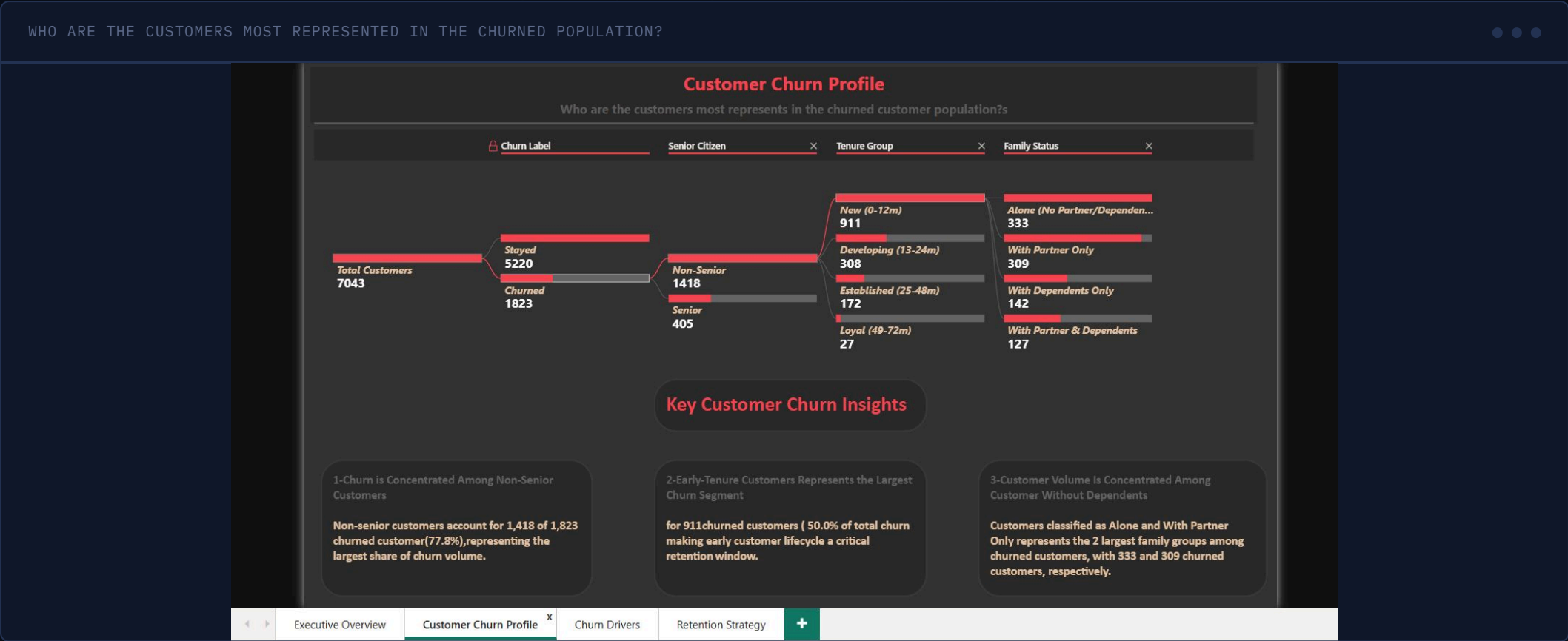
These questions guided the next stage of analysis.

Churn is concentrated among early-tenure customers.

This creates a critical retention window during the first year of the customer relationship.

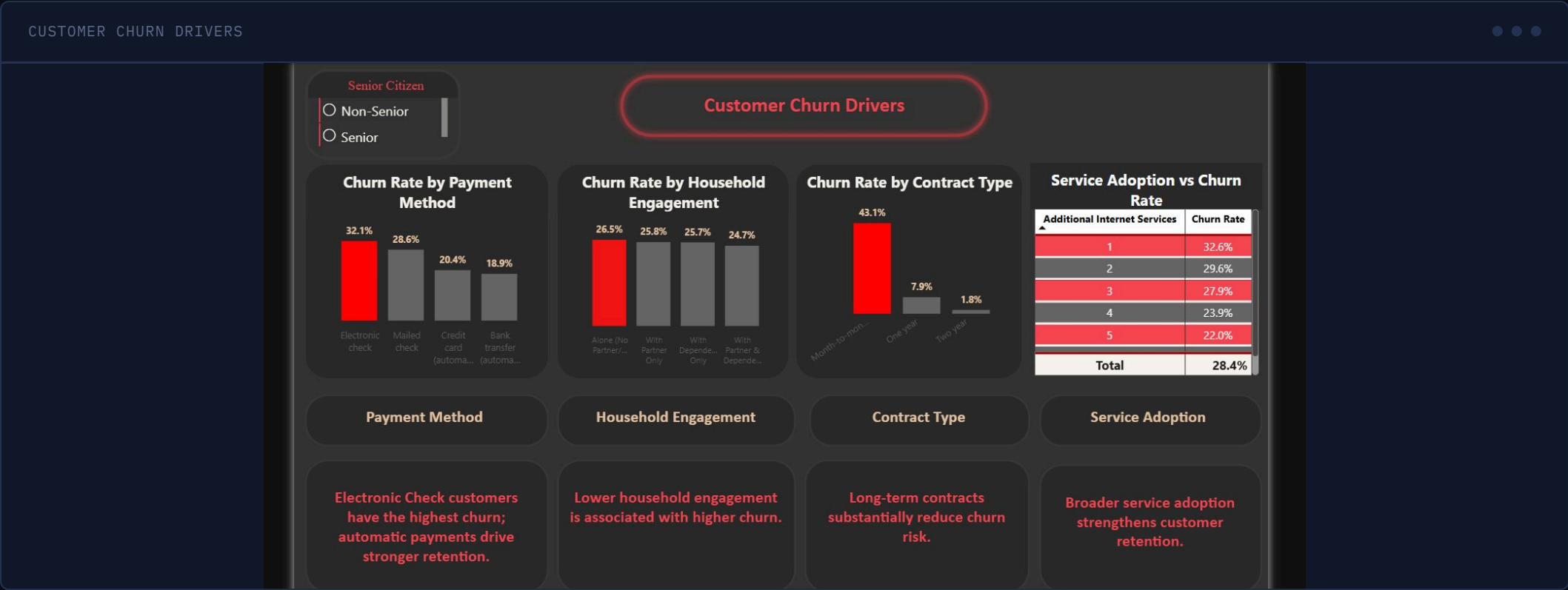


Customer Churn Profile



KEY FINDING Non-senior customers account for 1,418 of 1,823 churned customers (77.8%) — the largest share of churn volume. Early-tenure customers represent the largest churn segment: 911 churned customers, 50% of total churn, making the early customer lifecycle a critical retention window.

Customer Churn Drivers



Four patterns worth acting on.

01 — Early Tenure First-year exposure First-year customers represent the largest churn exposure.	02 — Contract Month-to-month risk Month-to-month contracts carry the highest churn risk.
03 — Payment Electronic check Electronic check customers show the highest churn rate among payment methods.	04 — Service Adoption Engagement effect Churn declines as customers adopt more additional services.

WHAT THIS MEANS FOR THE BUSINESS

Retention efforts should prioritize customers during their first year — especially those on month-to-month contracts and high-risk payment methods — while increasing service engagement.

High-Risk Customer Segments

Key risk metrics identified to guide targeted retention initiatives.

HIGHEST RISK SEGMENT First-year customers — 63.1% of churn occurs within the first year. New (0–12m)	HIGHEST RISK CONTRACT Highest churn among all contract types. Month-to-month
HIGHEST PAYMENT RISK Highest payment-related churn rate. Electronic check	STRATEGIC OPPORTUNITY More adopted services → lower churn. Expand Service Bundles

Action Plan & Business Impact

Data-driven strategy recommendations mapped to business outcomes.

DATA-DRIVEN RETENTION STRATEGY		
● Critical	Protect first-year customers Largest churn segment	
● Critical	Promote long-term contracts Strongest behavioral driver	
● Medium	Launch loyalty campaigns Improve retention	

EXPECTED BUSINESS IMPACT		
First-Year Program	Lower early churn	
Long-Term Contracts	Higher retention	
Service Bundles	Higher customer engagement	

Executive Recommendations

Summary of strategic imperatives and business priorities.

EXECUTIVE SUMMARY

- ✓ Increase Autopay adoption to reduce payment friction
- ✓ Focus retention efforts on the first year of the customer lifecycle
- ✓ Expand bundled services to strengthen customer engagement

BUSINESS PRIORITY

Contract Migration	● Critical
First-Year Retention	● Critical
Payment Optimization	● High

Four skills this project sharpened.

DATA

Structuring questions

Turning raw customer data into structured analytical questions.

POWER BI

Executive dashboards

Building an executive-level analytical dashboard.

SQL

Business queries

Using business-oriented queries to investigate customer behavior.

BUSINESS THINKING

Insight → action

Translating patterns into retention actions and business recommendations.

Data → **Insight** → Decision